

Experience

Independent Researcher & Open-Source Developer <i>a9l.im</i>	Feb 2026 - Present
Conduct independent interpretability and alignment research from hypothesis through experiments and writeup; publish open-source LLM tooling.	

Education

University of California, San Diego <i>B.S. Mathematics - GPA 3.75 - completed early</i>	Sep 2023 - Mar 2026
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Selected Research

Saklas <i>Mechanistic interpretability workbench - Python, PyTorch</i>	Apr 2026 - Present
- Built a local interpretability workbench compatible with most open source LLMs and both CUDA and MPS. - Implemented curved manifold extraction from sets of concepts. - Implemented Jacobian-lens readouts and nonstandard roles in chat templates. - Published a Python API, dashboard, and OpenAI/Ollama-compatible server	

Introspection via Kaomoji <i>Emotional concept representation study - Python</i>	Apr 2026 - Jun 2026
- Investigated correspondence between kaomoji emitted by LLMs to prompt emotional categories. - Found that the kaomoji could predict the emotional category with 80.6% accuracy on Gemma-4-31b-it. - Aligned hidden states across emotional category to analyze common geometry. - Built data collection tooling and published the resulting corpus.	

Elapsed-Time Encoding in LLMs <i>Ongoing time encoding study - Python, PyTorch</i>	Jun 2026 - Present
- Investigated the internal representation of elapsed time within an LLM over a conversation. - Showed elapsed time is encoded and linearly proportional to context length using probes.	

Neuralese as a Conlang <i>Ongoing abstract-CoT analysis study - Python, PyTorch</i>	Jul 2026 - Present
- Implemented Abstract-CoT to investigate if abstract tokens could be analyzed as a compositional language. - A preliminary run on a small model did not successfully learn to use abstract tokens to reason.	

Block Sparse Crosscoders <i>Ongoing cross-layer representation geometry study - Python, PyTorch</i>	Jul 2026 - Present
Synthesized recent work on block sparse autoencoders and sparse crosscoders to extract cross-layer multi-dimensional features.	

Selected Engineering

Geon <i>Relativistic N-body simulator - JavaScript, WebGPU</i>	Feb 2026 - Present
- Built a WebGPU simulation that models N-body physics with charged rotating massive particles. - Implemented finite-speed force propagation and Einstein-Infeld-Hoffmann dynamics. - Implemented force-coupled scalar fields with self-gravity.	

Ogdoad <i>Research library for recreational mathematics - Rust, Python</i>	Jun 2026 - Present
- Built a Rust library to explore Clifford algebras over the surreal numbers and Conway’s combinatorial games. - Implemented Python tooling and designed a purpose-built language to investigate open problems.	

Other Projects <i>Browser simulations and agentic tooling - JavaScript, WebGPU</i>	Feb 2026 - Present
Coding agent infrastructure. Gaslamp, a cross-agent CLI reviewer for Claude Code and Codex, and Dagwood, a version-controlled DAG-based task tracker with an MCP. Interactive simulations. Built seven more browser projects across physics, finance, biology, political science, and comparative religion using JS and WebGPU.	

Technical Skills

Research/ML: mechanistic interpretability; Python, PyTorch, NumPy/SciPy, CUDA, MPS.

Engineering: Java, Rust, JavaScript, WebGL, Git.s